

## TRIPOLI, Lebanon

WMO: 401030

Lat: 34.450N

Lon: 35.800E

Elev: 16

StdP: 14.69

Time Zone: 2.00 (LBN)

Period: 99-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 42.1       | 44.3 | 29.8                          | 23.9 | 48.7 | 33.7 | 28.3 | 49.9 | 22.1                  | 58.8 | 19.2 | 56.5 | 2.9                   | 160  | 0.327 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 13.0                   | 89.8            | 77.1 | 88.7 | 76.9 | 87.7 | 76.6 | 81.6                | 86.3 | 80.0 | 85.7 | 78.6 | 85.4 | 7.4                  | 280  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 80.3                            | 157.5 | 85.4 | 78.5 | 147.8 | 84.3 | 76.6 | 138.5 | 83.4 | 45.5          | 86.3 | 43.7 | 85.6 | 42.2 | 85.6 | 85.5              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| 17.7              | 14.4 | 12.3 | 37.7                       | 94.0 | 3.8                | 2.2 | 34.9                                               | 95.6 | 32.7       | 96.9 | 30.6       | 98.2 | 27.9       | 99.8 |    |    |
|                   |      |      | 34.7                       | 81.3 | 4.2                | 2.3 | 31.7                                               | 83.0 | 29.2       | 84.3 | 26.8       | 85.6 | 23.8       | 87.3 |    |    |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 68.2   | 54.6  | 56.1  | 59.6  | 64.2  | 70.3  | 76.2  | 80.2  | 81.7  | 79.3  | 73.3  | 65.0  | 57.4 |
|                                                                                  | DBStd       | 10.19  | 4.20  | 4.11  | 4.38  | 3.99  | 3.73  | 2.71  | 1.92  | 1.54  | 2.34  | 3.94  | 4.33  | 4.26 |
|                                                                                  | HDD50       | 24     | 12    | 6     | 1     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 5    |
|                                                                                  | HDD65       | 1101   | 323   | 250   | 175   | 60    | 3     | 0     | 0     | 0     | 0     | 2     | 51    | 237  |
|                                                                                  | CDD50       | 6678   | 155   | 176   | 300   | 427   | 628   | 785   | 938   | 983   | 878   | 723   | 451   | 234  |
|                                                                                  | CDD65       | 2280   | 0     | 0     | 9     | 38    | 166   | 335   | 473   | 518   | 428   | 260   | 52    | 1    |
|                                                                                  | CDH74       | 21478  | 0     | 1     | 49    | 173   | 896   | 2800  | 5339  | 6263  | 4231  | 1577  | 147   | 2    |
|                                                                                  | CDH80       | 6882   | 0     | 0     | 8     | 30    | 146   | 625   | 1951  | 2489  | 1349  | 278   | 6     | 0    |
| Wind                                                                             | WSAvg       | 4.7    | 4.8   | 5.0   | 5.2   | 5.1   | 4.9   | 5.3   | 5.3   | 5.1   | 4.8   | 3.7   | 3.4   | 4.1  |
| Precipitation                                                                    | PrecAvg     | 30.8   | 6.7   | 5.1   | 4.1   | 1.9   | 0.7   | 0.2   | 0.0   | 0.0   | 0.4   | 2.5   | 4.4   | 5.6  |
|                                                                                  | PrecMax     | 43.9   | 12.4  | 14.4  | 13.7  | 5.4   | 1.6   | 0.9   | 0.4   | 0.2   | 2.1   | 7.2   | 11.0  | 10.9 |
|                                                                                  | PrecMin     | 18.3   | 1.4   | 1.4   | 0.6   | 0.5   | 0.2   | 0.1   | 0.0   | 0.0   | 0.0   | 0.2   | 0.2   | 1.4  |
|                                                                                  | PrecStd     | 6.7    | 2.8   | 2.9   | 2.7   | 1.3   | 0.4   | 0.2   | 0.1   | 0.0   | 0.6   | 2.1   | 2.9   | 2.9  |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 69.2  | 71.7  | 79.6  | 83.7  | 88.1  | 89.4  | 91.4  | 91.8  | 90.5  | 88.6  | 80.7  | 71.9 |
|                                                                                  |             | MCWB   | 58.0  | 60.5  | 62.3  | 64.9  | 67.8  | 75.0  | 78.4  | 78.2  | 75.8  | 71.4  | 66.1  | 60.0 |
|                                                                                  | 2%          | DB     | 66.0  | 68.3  | 73.3  | 78.5  | 83.7  | 86.2  | 89.3  | 89.8  | 88.5  | 85.0  | 77.7  | 68.9 |
|                                                                                  |             | MCWB   | 57.1  | 58.9  | 61.1  | 65.1  | 68.8  | 74.6  | 77.9  | 77.8  | 75.8  | 71.2  | 64.4  | 59.0 |
|                                                                                  | 5%          | DB     | 64.1  | 66.1  | 69.8  | 74.9  | 80.7  | 84.7  | 88.1  | 88.8  | 87.1  | 82.9  | 75.3  | 66.9 |
|                                                                                  |             | MCWB   | 56.1  | 57.6  | 61.2  | 64.2  | 69.4  | 74.2  | 77.4  | 77.6  | 75.1  | 70.8  | 63.9  | 57.8 |
|                                                                                  | 10%         | DB     | 62.2  | 64.0  | 67.3  | 72.4  | 78.5  | 83.5  | 87.0  | 87.9  | 85.9  | 81.3  | 73.0  | 65.0 |
|                                                                                  |             | MCWB   | 54.8  | 56.3  | 59.3  | 63.2  | 68.9  | 73.7  | 76.9  | 77.3  | 74.5  | 70.2  | 62.7  | 56.8 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 60.9  | 62.3  | 65.9  | 69.0  | 74.7  | 79.2  | 82.9  | 83.9  | 81.4  | 77.1  | 68.9  | 62.9 |
|                                                                                  |             | MCDB   | 65.2  | 68.6  | 73.2  | 78.1  | 79.7  | 83.7  | 87.4  | 87.7  | 86.0  | 82.0  | 77.3  | 69.3 |
|                                                                                  | 2%          | WB     | 58.4  | 60.2  | 63.6  | 66.7  | 72.3  | 77.4  | 81.4  | 82.1  | 78.6  | 74.3  | 67.0  | 60.5 |
|                                                                                  |             | MCDB   | 64.2  | 66.9  | 70.2  | 74.4  | 79.4  | 83.6  | 86.3  | 86.5  | 85.5  | 81.8  | 75.1  | 66.5 |
|                                                                                  | 5%          | WB     | 56.8  | 58.4  | 61.8  | 65.3  | 70.8  | 76.3  | 80.0  | 80.4  | 77.1  | 72.8  | 65.4  | 59.0 |
|                                                                                  |             | MCDB   | 62.8  | 65.1  | 68.5  | 72.5  | 78.3  | 83.0  | 85.6  | 85.7  | 85.0  | 80.8  | 73.3  | 64.8 |
|                                                                                  | 10%         | WB     | 55.3  | 56.8  | 60.2  | 64.0  | 69.3  | 75.0  | 78.5  | 78.8  | 75.8  | 71.2  | 63.7  | 57.6 |
|                                                                                  |             | MCDB   | 61.4  | 63.4  | 66.7  | 71.1  | 77.1  | 82.0  | 85.1  | 86.1  | 84.3  | 79.6  | 71.3  | 63.7 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 12.9  | 13.7  | 14.5  | 14.9  | 15.0  | 13.7  | 13.4  | 13.0  | 13.6  | 15.3  | 15.6  | 13.9 |
|                                                                                  |             | MCDBR  | 15.3  | 16.9  | 18.6  | 19.4  | 18.7  | 15.5  | 14.7  | 14.0  | 14.9  | 16.6  | 17.9  | 16.2 |
|                                                                                  |             | MCWBR  | 9.8   | 10.9  | 11.6  | 11.4  | 11.4  | 9.9   | 9.0   | 8.3   | 8.9   | 9.8   | 10.3  | 9.8  |
|                                                                                  | 5% WB       | MCDBR  | 14.5  | 16.2  | 16.8  | 17.3  | 16.1  | 14.9  | 13.9  | 13.1  | 14.0  | 14.4  | 16.2  | 14.6 |
|                                                                                  | MCWBR       | 9.8    | 11.1  | 11.5  | 11.4  | 11.1  | 10.3  | 9.6   | 8.7   | 9.6   | 9.8   | 10.5  | 9.7   |      |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.379  | 0.411 | 0.454 | 0.504 | 0.496 | 0.437 | 0.452 | 0.461 | 0.441 | 0.452 | 0.399 | 0.384 |      |
|                                                                                  | taud        | 2.308  | 2.201 | 2.078 | 1.956 | 2.029 | 2.247 | 2.216 | 2.188 | 2.235 | 2.169 | 2.288 | 2.304 |      |
|                                                                                  | Ebn at Noon | 257    | 261   | 259   | 251   | 255   | 270   | 265   | 260   | 260   | 244   | 247   | 247   |      |
|                                                                                  | Edh at Noon | 33     | 40    | 49    | 58    | 55    | 44    | 45    | 46    | 42    | 41    | 33    | 31    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 777    | 1054  | 1519  | 1907  | 2305  | 2565  | 2510  | 2266  | 1901  | 1380  | 982   | 732   |      |
|                                                                                  | RadStd      | 65     | 105   | 88    | 124   | 118   | 52    | 37    | 45    | 59    | 69    | 69    | 70    |      |

## Historical Trends

|      |                         | DBAvg | Heating |        |       | Cooling |       |     | Degree-Days |       |       |       |
|------|-------------------------|-------|---------|--------|-------|---------|-------|-----|-------------|-------|-------|-------|
|      |                         |       | 99% DB  | 99% DP | 1% DB | 1% WB   | 1% DP |     | HDD50       | HDD65 | CDD50 | CDD65 |
|      |                         | (d)   | (e)     | (f)    | (g)   | (h)     | (i)   | (j) | (k)         | (l)   | (m)   |       |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A   | N/A   |       |
| (47) | Regional (13 neighbors) | +0.54 | N/A     | -2.39  | N/A   | N/A     | N/A   | N/A | N/A         | +311  | +198  |       |

Nomenclature: See separate page